CS5234 Mock Quiz 2

Lectures 5, 7–10 · open book · 50 minutes · 20 points 第 5、7–10 讲 · 开卷 · 50 分钟 · 共 20 分

Questions 1–4: one correct option each (1 point). Questions 5–6: show your reasoning; an essay question left completely blank receives 1 mark. log denotes the natural log unless stated.

第 1–4 题单选，每题 1 分。第 5–6 题需写出推理；完全空白的大题给 1 分。log 默认为自然对数。

1 [1 pt]

A stream of updates (i, Δ) with arbitrary integer Δ (frequencies may become negative). You want, for every i, an estimate with $|\tilde{x}_i - x_i| \le \varepsilon \|x\|_2$. Which sketch should you use?

一串更新 (i, Δ)，Δ 为任意整数（频率可以变成负数）。要对每个 i 给出估计，满足 $|\tilde{x}_i - x_i| \le \varepsilon \|x\|_2$。应选哪个 sketch?

(A) Misra-Gries

(B) Count-Min

(C) Count Sketch

(D) KMV

2 [1 pt]

In the KMV distinct-elements algorithm (output tm/X where X is the t-th smallest hash value), what is the weakest independence of the hash function that the lecture's analysis needs?

KMV 不同元素算法（输出 tm/X，X 为第 t 小的哈希值）中，课上的分析最少需要哈希函数满足哪种独立性?

(A) Fully (mutually) independent 完全独立

(B) 4-wise independent 4-wise 独立

(C) Pairwise independent 两两独立

(D) No independence is needed 不需要任何独立性

3 [1 pt]

The Morris counter increments C with probability 2^{-C} on each 1 and outputs $2^C - 1$. Let A be the number of 1s. Which statement is true?

Morris 计数器每来一个 1 以概率 2^{-C} 令 C 加 1，输出 $2^C - 1$。设 A 为 1 的个数。哪项正确?

(A) $\mathbb{E}[2^C] = A$. $\mathbb{E}[2^C] = A$。

(B) $\mathrm{Var}(2^C) = \frac{A(A-1)}{2}$, so averaging $\frac{10}{\varepsilon^2}$ independent copies gives a $(1 \pm \varepsilon)$-estimate with probability ≥ 0.9. $\mathrm{Var}(2^C) = \frac{A(A-1)}{2}$，所以平均 $\frac{10}{\varepsilon^2}$ 份独立副本能以 ≥ 0.9 的概率得到 $(1 \pm \varepsilon)$ 估计。

(C) Each copy needs $\Theta(\log n)$ bits, the same as an exact counter. 每份需要 $\Theta(\log n)$ 位，和精确计数一样。

(D) The analysis needs a pairwise-independent hash function. 分析需要两两独立的哈希函数。

4 [1 pt]

In an edge stream, the algorithm keeps H and adds an arriving edge iff it does not create a cycle of length ≤ 4 in H (i.e. $k = 2$). Which is true?

边流中，算法维护 H，到达的边当且仅当不在 H 中形成长度 ≤ 4 的环时才加入（即 $k = 2$）。哪项正确?

(A) H is a 3-spanner of G with $O(n^{3/2})$ edges. H 是 G 的 3-spanner，有 $O(n^{3/2})$ 条边。

(B) H is a 4-spanner of G with $O(n\sqrt{n}\log n)$ edges. H 是 G 的 4-spanner，有 $O(n\sqrt{n}\log n)$ 条边。

(C) H is a 3-spanner, but it can have $\Theta(n^2)$ edges. H 是 3-spanner，但可能有 $\Theta(n^2)$ 条边。

(D) H preserves all distances exactly. H 精确保持所有距离。

5 [9 pts]

Two vectors $x, y \in \mathbb{Z}^m$ start at 0. A single stream interleaves updates (x, i, Δ) meaning $x_i \mathrel{+}= \Delta$ and (y, i, Δ) meaning $y_i \mathrel{+}= \Delta$, with arbitrary integer Δ (turnstile). All entries stay bounded by poly(n). Design a streaming algorithm that outputs $D = \|x - y\|_2^2$ within $(1 \pm \varepsilon)$ with probability $1 - \delta$.

两个向量 $x, y \in \mathbb{Z}^m$ 初始为 0。同一条流里交替出现更新 (x, i, Δ) 表示 $x_i \mathrel{+}= \Delta$，(y, i, Δ) 表示 $y_i \mathrel{+}= \Delta$，Δ 为任意整数（turnstile）。所有分量绝对值不超过 poly(n)。设计流算法，以概率 $1-\delta$ 输出 $D = \|x-y\|_2^2$ 的 $(1 \pm \varepsilon)$ 近似。

(a, 2 pts) Describe one basic estimator: what is stored and how each update is processed. Why must x and y use the *same* random signs?

描述一个基本估计量：存什么、每条更新怎么处理。为什么 x 和 y 必须用同一组随机符号？

(b, 4 pts) Show $\mathbb{E}[Z^2] = D$ and $\mathrm{Var}[Z^2] \le 2D^2$. State exactly where you use 4-wise independence.

证明 $\mathbb{E}[Z^2] = D$ 且 $\mathrm{Var}[Z^2] \le 2D^2$，并指出在哪里用到 4-wise 独立。

(c, 3 pts) Choose the number of copies and give the total space in bits.

确定副本数，给出总空间（位）。

6 [7 pts]

In problem DUP, the stream is a sequence of numbers from $[m]$ and the algorithm must say whether some number appears at least twice. You may use: any randomized one-way protocol for Index on m bits with error $\le 1/3$ needs $\Omega(m)$ bits.

问题 DUP：流是一串 $[m]$ 中的数，算法要判断是否有某个数出现至少两次。可直接使用：m 位 Index 问题的任何错误率 $\le 1/3$ 的随机单向协议都需要 $\Omega(m)$ 位。

(a, 4 pts) Show that every one-pass streaming algorithm for DUP that succeeds with probability $\ge 2/3$ uses $\Omega(m)$ bits of memory.

证明：任何成功率 $\ge 2/3$ 的单遍 DUP 流算法都需要 $\Omega(m)$ 位内存。

(b, 1 pt) Why is it fine that the streaming algorithm is randomized?

为什么流算法是随机的也没关系？

(c, 2 pts) Use the same idea to show that computing the *exact* number of distinct elements DE needs $\Omega(m)$ bits.

用同样的思路证明：精确计算不同元素个数 DE 需要 $\Omega(m)$ 位。